

TEZPUR UNIVERSITY

Mid-Semester Examination, Autumn 2023

CS621: MOBILE COMPUTING

Full Marks: 40

Time: 2hrs

Figures in the right hand indicate marks for the individual questions

4×3=12

1. Distinguish the following (any 4)

- a. Hidden Terminal problem vs. Exposed Terminal problem
- b. Reporting Cell vs. Location Area based Location Management
- c. Adaptive Guard Channels vs. Queuing Handoff Calls
- d. Fixed Channel Assignment vs. Dynamic Channel Assignment
- e. Sequential Paging vs. Group Sequential Paging
- f. Hard Handoff vs. Soft handoff

2. Suppose, the minimum SNR required by cell phone receiver in a cellular network is greater than or equal to 24dB. What will be the reuse factor if path loss exponent (α) is given as 4? [Note: Use the relationship between Co-channel separation distance D, reuse factor K and cell radius R. Also assume that the user is located in a corner of a hexagonal cell and only first tier of interfering cells (i.e. 6 interfering cells).]

3. Suppose the probabilities of a cell phone user to be in cells A1, A2, A3, A4 and A5 are given as 0.15, 0.10, 0.25, 0.30 and 0.20 respectively. Compute the **expected paging delay** if sequential paging is used to locate the user.

4. List out the key technological factors in achieving higher data rates in newer generations of WiFi. Why does carrier sensing fail in Wireless Network?

5. What are the purposes of the numbers - MSISDN, IMSI, TMSI & IMEI used in GSM?

6. What is the need of Timing Advance in GSM? Suppose a mobile device has received an update of **Timing Advance** as 14 from the BTS (in GSM terminology), estimate the distance of the mobile device from the BTS in km.

7. Suppose your mobile is registered in Chennai and you are currently in Delhi (and updated your location). What happens when your friend in Chennai calls you (that means, how call will be delivered to your current location)? Assume that each metro city is under a single MSC. Use different components of Cellular Networks applied in location management.

8. What do you mean by location management in cellular system?

9. Match the entries in Column A with the **most appropriate** entries in Column B

Column A

Column B

(a) Locator Receiver

(b) Handoff calls

(c) Beamforming

(d) Profile based location Management

(e) Always Update

(f) SSID

(g) Passive Scanning

(h) Walsh Code

(i) Beacon frame

(ii) Paging cost

(iii) Mobility Pattern of users

(iv) CDMA

(v) Network Controlled Handoff

(vi) Macro Cell

(vii) MIMO

(viii) Extended Service Set (ESS)

(ix) Forced Termination Probability